



VERIFICATION REPORT FOR “AES SAURASHTRA WINDFARMS”



Document Prepared By Earthood Services Private Limited

Project Title	AES Saurashtra Windfarms
Version	01.1
Report ID	VCS.VER.22.45

Report Title	Verification report for “AES Saurashtra Windfarms”
Client	Tata Power Renewable Energy Limited
Pages	45
Date of Issue	09/11/2022
Prepared By	Earthood Services Private Limited
Contact	409-410, Tower B4, Spaze I-Tech Park, Sector 49, Gurugram 122018, INDIA Tel: +91 124 4204599

	Fax: +91 124 4204599	
	Website: www.earthood.in	Email: info@earthood.in
Approved By	Dr. Kaviraj Singh	
Work Carried Out By	Harsh Raval – Team Leader, Local Expert, Meth Expert, and Technical Expert (TA1.2)	
	Muskan Chawla – Trainee Verifier	
	Vikas Bankar – Technical Reviewer and Technical Expert (TA 1.2)	

Summary:

Earthood Services Private Limited (hereafter referred to as ESPL) has been contracted by Tata Power Renewable Energy Limited to conduct the verification of the project – “AES Saurashtra Windfarms”, (VCS ID-1039) with regards to the relevant requirements of VCS programme guidelines and standard (VCS standard version 4.3)^{/05/} and VCS program guide version 4.3^{/06/}. Relevant requirements of the UNFCCC for CDM project activities, as well as criteria for consistent project operations, monitoring, and reporting have been applied for verification.

The monitoring period covered under this verification is from 01/09/2018 to 06/06/2021.

The verification includes confirming the implementation of the monitoring plan of the registered PD (VCS ID 1039) and the application of the monitored methodology as per ACM0002 version 12.3.0, EB 66 (Consolidated baseline methodology for grid-connected electricity generation from renewable sources).

The project activity involves electricity generation by wind power project of 39.2 MW wind farm in the Indian state of Gujarat and supplying the generated electricity to the state electricity utility. The project is a renewable energy generation activity that results in the reduction of greenhouse gas (GHG) emissions that are real, measurable, and verifiable and plays a beneficial role in the mitigation of climate change.

A risk-based approach has been followed to perform this verification. During verification 01 Corrective Action Request (CARs), 00 Forward Action Request (FARs), and 05 Clarification Request (CLs) were raised and successfully closed.

The review of the project description, monitoring report, and additional documents related to baseline and monitoring methodology; the subsequent background investigation, telephonic interviews, and stakeholders have provided ESPL with sufficient evidence to validate the fulfillment of the stated criteria. ESPL confirms that the project is implemented in accordance with the registered PD^{/01/} and MR^{/02/}. The monitoring system is in place and the emission reductions are calculated without material misstatements. Our opinion relates to the project's GHG emissions, and the resulting GHG emission reductions reported and related to the valid and registered project baseline and monitoring and its associated documents. Based on the

information seen and evaluated we confirm that the emission reductions from the project activity, 'AES Saurashtra Windfarms' during the period 01/09/2018 to 06/06/2021 amounts to 162,589 tons of CO_{2e} with displacing 176,250.06 MWh of electricity from fossil fuel dominating energy grid with electricity generated using wind energy system.

1	Introduction	6
1.1	Objective	6
1.2	Scope and Criteria	6
1.3	Level of Assurance.....	6
1.4	Summary Description of the Project	7
2	Verification Process	7
2.1	Method and Criteria.....	7
2.2	Document Review	8
2.3	Interviews.....	9
2.4	Site Inspections.....	10
2.5	Resolution of Findings	13
2.5.1	Forward Action Requests	14
2.6	Eligibility for Validation Activities	14
3	Validation Findings	14
3.1	Participation under Other GHG Programs	14
3.2	Methodology Deviations.....	14
3.3	Project Description Deviations.....	15
3.4	Grouped Project	15
4	Verification Findings	15
4.1	Project Implementation Status	15
4.2	Safeguards	18
4.2.1	No Net Harm	18
4.2.2	Local Stakeholder Consultation	18
4.3	AFOLU-Specific Safeguards	18
4.4	Accuracy of GHG Emission Reduction and Removal Calculations	18
4.5	Quality of Evidence to Determine GHG Emission Reductions and Removals	20
4.6	Non-Permanence Risk Analysis.....	32
5	Verification conclusion	33
	APPENDIX 1: Document References	35
	Appendix 2: Abbreviations	37
	Appendix 3: Findings Overview	39

Appendix 4: Competency Statements

44

1 INTRODUCTION

1.1 Objective

Earthood Services Private Limited (ESPL) has been contracted by Tata Power Renewable Energy Limited, to undertake the verification of the renewable energy project titled “AES Saurashtra Windfarms” (VCS ID-1039). The verifiers have reviewed the GHG data collected to date for the monitoring period from 01/09/2018 to 06/06/2021 covered in this verification.

The objective of this verification is a thorough and independent assessment of registered project activities against the applicable VCS requirements by the VVB. The verification process shall determine whether the proposed project activity complies with the requirement of the latest VCS guidelines, applicability conditions of the selected methodology, relevant host country regulations, and guidance issued by the VCS Board.

1.2 Scope and Criteria

The scope of this verification is to assess the claims and assumptions made in the VCS monitoring report against the VCS criteria, including but not limited to, VCS standard V4.3, applied methodology, and other relevant rules and requirements established for VCS project activities.

The verification is not meant to provide any consulting to the project participants. However, stated requests for clarification and/or correction actions requested may have provided inputs for improvement of the project design.

1.3 Level of Assurance

☒ Reasonable level of assurance

☐ limited level of assurance

The level of assurance of the verification report falls under reasonable assurance engagement as selected by the client. Reasonable assurance is a high level of assurance regarding material misstatement, but not an absolute one.

Reasonable assurance includes the understanding that there is a remote likelihood that material misstatements will not be prevented or detected on a timely basis. To achieve reasonable assurance, the auditor needs to obtain sufficient appropriate audit evidence to reduce audit risk to an acceptably low level. This means that there is some uncertainty arising from the use of sampling since it is possible that a material misstatement will be missed.

The evidence used to achieve a reasonable level of assurance is specified in sections 2.3 and 2.4 of this report.

1.4 Summary Description of the Project

The project activity is a wind power project of 39.2 MW developed by Tata Power Renewable Energy Limited in the Indian state of Gujarat. The project activity consists of 49 E- 53 WTGs of 800kW each totaling a total generation capacity of 39.2 MW.

The project aims to generate cleaner electricity using wind power project and displace the amount of electricity generated using the fossil fuel-based power plant. The electricity generated is supplied to the state electricity utility.

A remote audit was conducted on 18/07/2022, PP representatives were interviewed through a video call and it was checked that the current monitoring period from 01/09/2018 to 06/06/2021 involves a supply of 176,250.06 MWh of electricity to the grid and the total emission reductions of 162,589 tCO₂e. The details of the alternative means of verification can be found in the section 2.3 and section 2.4 of this report.

The start date of the project activity is 07/06/2011 which is the date of actual commercial operation of the project activity. The crediting period of the project is from 07/06/2011 to 06/06/2021, the same is verified using the registered PD^{/01/}, validation report^{/04/}, and the remote audit^{/10/}.

The project is based on the methodology ACM0002, version 12.3.0, EB 66, “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”^{/08/}. Based on the assessment of the documents, the assessment team can confirm that the project activity is fully functional and implemented as described in the registered PD^{/01/} and MR^{/02/}.

2 VERIFICATION PROCESS

The registered VCS project is undergoing fourth verification, the approach adopted to ensure the quality of emissions reductions is described in the following sections.

2.1 Method and Criteria

The verification approach consists of two phases:

In the first phase, ESPL completed a strategic review and risk assessment of the projects activities and processes to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leaking if relevant.

- Protocols used to estimate or measure GHG emissions from these sources.
- Collection and handling of data.
- Controls the collection and handling of data.
- Means of verifying reported data.
- Compilation of the verification Report.

At the end of this phase, ESPL produced a Verification Checklist which, based on the risk assessment of the parameters and data collection and handling processes for each of those parameters, describes the verification approach and the sampling plan.

In the second phase, ESPL verified the implementation of the monitoring plan and the data presented in VCS MR^{/02/} for the period in question. This involves telephonic interviews of project proponent representatives and a desk review of the Monitoring Report. This verification report describes the findings of this assessment.

2.2 Document Review

The verification is performed primarily as a document review of the registered VCS PD^{/01/} and MR^{/02/}. The review is performed by the assessment team using dedicated protocols. The assessment team cross-checks the information provided in the documents and information from a source other than those used, if available, and conducts independent background investigations. Earthood conducts a desk review as under:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures;

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emissions reduction.

2.3 Interviews

In accordance with section 4.1.2 of the VCS standard, v4.3, the VVB is required to reach a reasonable level of assurance. If reasonable assurance can be achieved, then it is not mandatory for VVB to conduct the on-site visit for verification. However, a remote audit was conducted on 18/07/2022 i.e., the representatives of the PP were interviewed remotely, personnel responsible for the monitoring of the project activity, data collection and management, and QA/QC procedure. The details of the people interviewed are mentioned in the table below:

Interview with PP representatives:

S. No.	Name	Organization	Topic covered	Team Member
1.	Mr. Prashant Tiwari	Tata Power Renewable Energy Limited	Project implementation, Commissioning dates as per the requirements of VCS.	Harsh Raval Muskan Chawla
2.	Mr. Sumit Shukla	Plant Head, AES Saurashtra Windfarms	Reliability and accuracy of readings are considered for emission reduction calculations, calibration procedure Monitoring, and measuring system. Collection of measurements Observations of established practices for verification of monitoring parameters Ongoing Local stakeholders' consultation process	Harsh Raval Muskan Chawla
3.	Mr. Anil Kumar	Project Manager, EKI Energy Services Ltd.	Electricity generation records (Monthly Energy Statements, Invoices)Emission reduction calculation	Harsh Raval Muskan Chawla

Interview with Local stakeholders:

Name	Designation	Topic Discussed	Team Member
Mr. Bharat Karmur	Senior technician	Grievance Mechanism,	Harsh Raval Muskan Chawla
Mr. Aala Chavda	Senior technician	Project implementation,	Harsh Raval Muskan Chawla
Mr. Arvin Chudasama	Senior technician	Employment opportunities	Harsh Raval Muskan Chawla
Mr. Jaga Bhayiya	Senior technician	Grievance Mechanism,	Harsh Raval Muskan Chawla
Mr. Kishor Van	Technician	Project implementation,	Harsh Raval Muskan Chawla
Mr. Malde Varu	Supervisor	Employment opportunities	Harsh Raval Muskan Chawla

It was concluded that the local people were aware of the grievance mechanism and there were no negative comments from local stakeholders side. Instead, the local people appreciated the development and job created by the project activity in that area. Thus, a reasonable level of assurance was met.

2.4 Site Inspections

As discussed in the above section, a physical site inspection is not done for the current verification. However, to achieve a reasonable level of assurance, the assessment team has followed the alternative means to substantiate the verification criteria as described in the below table:

Assessment Criteria	Means of verification/ Source documents	Assessment opinion
Description of the project activity	Registered PD ^{/01/} and validation report ^{/04/} . Virtual interviews with site personnel and PPs	The information's with reference to project capacity, technology, plant equipment and commissioning dates as provided in section 1 of

Assessment Criteria	Means of verification/ Source documents	Assessment opinion
	representatives on 18/07/2022 ^{/10/} .	MR are found consistent with the documents.
Compliance of the project implementation with the registered project design documents	Monthly share certificates issued by GETCO^{/09/}. Geo coordinates of the project activity verified through google map Virtual interviews with the site personnel and PPs representatives on 18/07/2022^{/10/}.	Monthly share certificates issued by GETCO and indicated the following information: Identification of substation to which the project is connected, serial numbers of energy meters used for measurement, Capacity of project, and name of project participant. The location of the project is verified through GPS Map and found consistent with registered PD^{/01/}. All the information regarding the project implementation as discussed above is further verified through registered PD^{/01/} and found consistent.
Compliance of the registered monitoring plan with applied methodologies and standardized baseline	Virtual Interviews with site personnel and PPs representatives on 18/07/2022^{/10/}. Monthly share certificates issues by GETCO^{/09/}.	The organizational structure, responsibilities, and competencies of the personnel confirm through interviews. The frequency of monitoring of parameters listed under the approved monitoring plan is verified through GETCO Certificates. The method used for measuring recording, storing, aggregating, and reporting the data on monitored parameters is verified through PPA and virtual conservations with site personnel.

Assessment Criteria	Means of verification/ Source documents	Assessment opinion
		Procedure for data uncertainty, emergency preparedness, roles and responsibility, and operational and management structure is mentioned in the MR ^{/02/} is confirmed and found satisfactory.
Compliance with the calibration frequency requirements for measuring instruments	Calibration certificates of meters installed ^{/11/} Central Electricity Authority (Installation and Operation of Meters) Regulations ^{/14/}	Calibration frequency and energy meter specifications (Sr. No, make accuracy class) are verified through calibration certificates ^{/11/} are found to be consistent. Responsibility for calibration and maintenance of energy meters is solely under the control of GETCO this is verified through the respective PPA.
Assessment of data and calculation of emission reductions or net removals	Monthly Share certificates issued by GETCO ^{/09/} CEA CO ₂ Baseline database for the Indian Power Sector ^{/17/}	Monthly values of monitoring parameters used in ER calculation are verified through share certificates and cross-verified with the respective invoices. Methods, formulae, and emission factors for calculating baseline emissions have been followed in accordance with the applied methodology ^{/08/} .

It is noteworthy that no sampling plan for verification is applied as 100% data is verified for the current monitoring period. Most of the reference documents referred to by the assessment team (in the above table) are either issued by the state utility (GETCO) or an external government agency, hence deemed authentic.

Based on the above assessment it can be concluded that the assessment team has verified sufficient appropriate audit evidence, to reduce audit risk to an acceptably low level as required to achieve a reasonable level of assurance for the current verification.

2.5 Resolution of Findings

The objective of the step is to identify, discuss, and conclude the issues related to the monitoring, implementation, and operations of the registered project activity that could impair the capacity of the registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on the desk review and onsite assessment. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and nonconformities, which may be of the following types:

CAR (Corrective Action Request) is raised if one of the following occurs:

- a) Non-compliance with the monitoring plan, the methodology, or the standardized baseline is found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient.
- b) Modifications to the implementation, operation, and monitoring of the registered project activity have not been sufficiently documented by the project participants.
- c) Mistakes have been made in applying assumptions, data, or calculations of emission reductions that will impact the number of emission reductions.
- d) Changes to the key sustainable development indicators.
- e) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

A clarification request (CL) is raised if the information is insufficient or not clear enough to determine whether the applicable requirements have been met. All CARs and CLs raised by the Earthhood during verification shall be resolved before submitting a request for issuance.

FAR (Forward Action Request) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

A total of 05 CLs and 01 CAR were raised in the current verification. No FAR was raised. All the findings that are raised and communicated to project participant during the verification are included in Appendix 3. The section also includes the responses by the project participants and an assessment by the verification team if it was closed out or otherwise.

2.5.1 Forward Action Requests

The project activity is undergoing fourth verification. No FAR is raised in the current monitoring period.

Details of this can be checked in appendix 3 of this verification report.

2.6 Eligibility for Validation Activities

This part is not applicable for present verification, as ESPL holds the accreditation for the Validation of projects under this Sectoral scope.

3 VALIDATION FINDINGS

Not applicable.

3.1 Participation under Other GHG Programs

The project activity is CDM registered project, reference number- 5777.

The web link for the project is added below

<https://cdm.unfccc.int/Projects/DB/DNV-CUK1328700673.83/view>.

The project is also registered under VCS. (VCS ID- 1039)

The PP has submitted the declaration ¹⁰⁷¹ which states that the net GHG emission reductions generated by the project activity will not be used for compliance with any other emissions trading program or to meet binding limits on GHG emissions for the same monitoring period. The further Assessment team has checked the undertaking dated 13/07/2022 provided by the PP confirming that there would not be any double counting of GHG credits.

CL#01 was raised regarding the same and closed as the PP submitted the declaration.

3.2 Methodology Deviations

There is no deviation in the methodology during the current monitoring period.

3.3 Project Description Deviations

The PP has taken a permanent deviation in this monitoring plan. The government agency, GETCO/GEDA only provide the direct readings of net export, the values for the parameters $EG_{clustermeter,Export}$, $EG_{clustermeter,Import}$, $EG_{GETCO,Export}$ & $EG_{GETCO,Import}$ are not provided to the PP, which were early part of the monitoring parameters. These parameters are only monitored to calculate the net export by apportioning and it is not in control of the PP. therefore, the PP has requested a permanent deviation in the monitoring plan and these parameters will not be monitored.

The assessment team has verified the deviation taken. It is concluded that the deviation taken will not impact the applicable methodology, additionality or appropriateness of the baseline scenario.

3.4 Grouped Project

This is not applicable, as this is not a grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project activity involves the installation of WTGs to generate electricity from wind energy. The installed capacity of the project is 39.2 MW (49 E-53 machines of 800 kW each) in the Indian state of Gujarat. Since, this is the fourth verification, the commissioning, capacity, project implementation, grid connectivity, and technical specifications have been checked during the validation^{/04/} and previous verifications.

This proposed wind power project will reduce the GHG emissions generated by the current generation energy mix in India's Power Grid, which is dominated by fossil fuel-based grid-connected power plants. The Power generated through the proposed project activity will be supplied to the state utility under a long-term power purchase agreement, as verified using the registered PD^{/01/}.

The start date of the project activity is 07/06/2011, which is also the date of the actual commercial operation of the project activity. This is the fourth verification of the project activity from 01/09/2018 to 06/06/2021. The crediting period of the project activity is from 07/06/2011 to 06/06/2021. The project has delivered 176,250.06 MWh of electricity to the national grid which has resulted in the reduction of 162,589 tCO₂e as verified from the documents^{/02/, /03/}. The assessment team did a remote site visit and interviews in order to verify the status of the project implementation of the monitoring plan. It has also been verified as per the guidelines in ACM0002, version 12.3.0, 'Consolidated baseline methodology for grid-connected electricity generation from renewable sources'^{/08/} that the calculations for the GHG emission reductions are done in accordance with the aforesaid methodology.

The technical specification of the project activity equipment has been checked through the photographs^{/10/} of all the equipment installed at the site and are found to be consistent with the details mentioned under MR. The current status of the project activity is verified through generation data for project investors and WTGS and hence it is confirmed that the project is fully functioning.

The assessment team has reviewed the monthly share certificates^{/09/}, and confirmed that the WTGs are connected to the EIL substation at Bhogat. Hence, to calculate the net electricity exported to the grid by the WTGs of the project activity, the state electricity utility uses an apportioning procedure that has been correctly described in section 4.3 of the MR^{/02/} and in the PD^{/01/}. This was also verified by interviewing the staff at the site during the remote audit.

It is verified that share certificates are issued by state utility^{/09/} and the apportioning procedure is carried out by the state utility and the PP has no role in this calculation. The required monitoring system has been installed and is operational. The meters comply with appropriate quality standards applicable to the technology. The accuracy class of the meters installed for the project activity was verified against the registered PD^{/01/} and VCS MR^{/02/}.

Monitoring Procedure:

The WTGs are connected to various clusters at the site and each cluster is connected to a dedicated metering system. The monitoring of the data takes place at Cluster metering point and GETCO main meter at the EIL Bhogat substation. Meter reading is taken by the state utility (GEDA) on the monthly basis and accordingly, monthly share certificates are prepared. This was also verified by interviewing the staff at the site during remote audit^{/10/}.

The net electricity that the wind farm contributes to the grid is calculated by the Gujarat Energy Development Agency (GEDA) and certified by the State Load Dispatch Center (SLDC/GETCO) using the meter readings from the individual cluster meters and the main GETCO meter after accounting for transmission loss.

Values of the parameter “Net Quality of Electricity exported to the grid” i.e., $EG_{PJ,y}$ is directly sourced from the monthly Share certificates issued by GETCO^{/09/} that indicate the share of electricity for project activity received at the respective pooling sub-stations. The measurement results are cross-checked with records of share certificates, and it is in line with the applied methodology. Thus, this parameter is considered in emission reduction calculations.

The net electricity supplied to the grid by project activity is being measured continuously by energy meters located at the pooling substation and recorded on at least a monthly basis. This is in line with methodology and is accepted.

The VCS MR^{/02/} has been reviewed to check that the procedure for data uncertainty, emergency preparedness, roles and responsibility, and operational and management structure is mentioned in the MR. The monitoring plan completely describes all measures to be implemented for monitoring all parameters required. The monitoring plan described the positioning of the equipment. The calibration frequency of the meter is once in 3 years, which was found to be appropriate and is conformed through the calibration certificates^{/11/}.

The project proponent's contribution from the project activity towards sustainable development in accordance with to host country as explained below:

Social Well being:

The project will help in generating employment opportunities for professional, skilled, and unskilled people during different phases. The project activity will lead to the development of infrastructure.

Economic Well being:

The project is a clean technology investment in the region, which will help in reducing the demand-supply gap in the state.

Environmental Well being:

As the project activity is a clean technology, it will reduce the amount of NO_x, SO_x, and particulate matter emissions which will lead to an improvement in the air quality in the nearby area.

Technological Well being:

The project activity is a clean and environmentally safe technology, it will help will contributing more green energy to the grid system.

The assessment team concludes the following:

- There are no material discrepancies between project implementation and the project description provided in the registered PD^{/01/}
- The monitoring plan is implemented completely and the monitoring system (i.e., process and schedule for obtaining, recording, compiling, and analyzing the monitored data and parameters) is appropriate.
- There are no material discrepancies between the actual monitoring system, the monitoring plan set out in the project description, and the applied methodology^{/08/}.
- The GHG emission reductions or removals generated by the project have not been included in an emissions trading program or any other mechanism that includes GHG allowance trading^{/07/}.

The project activity is complying with indicators for sustainable development in the interim approval guidelines for Clean Development Mechanism (CDM) projects from India as discussed under section 1.11 of MR. In view of the information verified above the assessment team can conclude that the project has been implemented as described in the project description.

CL#03 was raised regarding the clarification of the crediting period. The CL was resolved subsequently.

4.2 Safeguards

4.2.1 No Net Harm

The project activity does not involve any negative environmental or socio-economic impacts.

There is no negative impact on any socio-economic conditions of the region due to the project activity. This project activity does not involve any negative environmental or socio-economic impacts, as the project activity involves the generation of power using wind energy which is a clean source of energy. Hence no mitigation measures are required.

4.2.2 Local Stakeholder Consultation

The project activity is undergoing fourth verification and the local stakeholder's consultation was appropriately conducted before validation as a way to inform the design of the project and maximize participation from stakeholders.

The details of the local stakeholder consultation, along with the concerns raised can be found in section 2.2 of the MR.

The project proponent has also established a mechanism for continuous stakeholder interactions, a dedicated email address ID tatapower@ethics-line.com and toll-free numbers 008001004382 and 0008001008277 have been provided to the local stakeholders to lodge their complaints, also it was confirmed that there were no comments received from the local stakeholders regarding the project activity. The detailed description of the same can be found in section 2.2 of MR.

CL#02 was raised regarding ongoing local stakeholder consultation and was successfully resolved.

4.3 AFOLU-Specific Safeguards

Since the project is a non-AFLOU project, hence not applicable.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

The calculations for the emission reductions are found to be correct. The details of the reported and verified values of all the parameters are listed in section 4.5 of this report.

This data has been verified as described in section 4.5 below. The formulae and method used to calculate the baseline emissions, project emissions and leakage are appropriate and in line with the approved methodology ACM0002 version 12.3.0 ^{/08/}.

The PP has calculated the grid emission factor as per the combined margin approach described in the 'Tool to calculate the emission factor for an electricity system', version 2.2.1, EB 63, Annex 19^{/15/}.

The grid emission factor has been calculated as the weighted average of OM and BM, and has been fixed ex-ante for the entire crediting period. The OM and BM have been obtained from a publicly available source i.e., "CO₂ Baseline Database for Indian Power sector", version 5 ^{/17/} published by the Central Electricity Authority, Ministry of Power, and Government of India. The OM has been determined as the average of the previous 3 years' values obtained from the CEA database^{/14/}.

The OM and BM have been calculated to be 1.0049 tCO₂/MWh and 0.6752 tCO₂/MWh respectively. The combined margin has been calculated to be 0.9225 tCO₂/MWh.

The formula used to calculate Emission reduction:

$$ER_y = BE_y - PE_y$$

Where,

BE_y = Baseline Emissions in year y (t CO_{2e}/yr)

PE_y = Project Emissions in year y (t CO_{2e}/yr)

For calculation of baseline emissions:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where,

BE_y = Baseline Emissions in year y (t CO_{2e}/yr)

EG_{PJ,y} = Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

EF_{grid,CM,y} = CO₂ emission factor of the grid in year y (t CO₂/yr)

Since the project activity is the installation of a new grid-connected renewable power plant the EG_{PJ,y} is calculated as :

$$EG_{PJ,y} = EG_{facility,y}$$

EG_{facility,y} = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$EG_{PJ,y} = 176,250.06 \text{ MWh}$$

$$EF_{grid,CM,y} = 0.9225 \text{ tCO}_2/\text{MWh}$$

$$BE_y = 176,250.06 * 0.9225 = 162,589 \text{ tCO}_2\text{e}$$

$$BE_y = 162,589 \text{ tCO}_2\text{e}$$

The emission reduction as

$$ER_y = BE_y - PE_y$$

$$PE_y = 0$$

$$\text{Thus, } ER_y = BE_y$$

$$ER_y = 162,589 \text{ tCO}_2\text{e}$$

The assessment team can confirm that the GHG emission reductions and removals have been quantified correctly in accordance with the project description and applied methodology.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All the data recorded comply with the registered PD^{/01/} and Monitoring Report^{/02/}. The assessment team has checked the monthly share certificates for electricity generated and supplied by project activity^{/09/} for the current monitoring period to verify the values of monitoring parameters reported in ER calculation sheet and found them to be consistent. Since certificates are prepared and issued by state utility based on the monthly joint meter readings by PP and state agency and also used for commercial invoice purposes, they are found to be reliable and authentic.

The monitoring of the project activity is found to be in accordance with the monitoring methodology described in ACM0002, Version 12.3.0^{/08/}. The monitoring mechanism is effective and reliable. During the remote audit, personnel involved at various levels of the operation of the project activity have been interviewed to confirm that the plant personnel is conscious of the importance of the monitoring activities.

The meters comply with appropriate quality standards applicable to the used technology. The accuracy class of the meters installed for the project activity was verified through the registered CDM PD^{/01/}, MR^{/02/}, and calibration certificates^{/11/} are found to be consistent.

The following parameter has been verified for the current monitoring period:

Parameters:

1. Net Quantity of Electricity exported to the grid y, $EG_{PJ,y}$ (MWh):

Means of verification	Criteria/ Requirements	Assessment/ Observation
	Measuring/Reading/Recording frequency	The parameter is calculated and recorded on monthly basis,

		however, the input values used to calculate the value of EG _{pj,y} are continuously monitored, hourly measured, and monthly recorded.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The measuring and reporting frequency is in line with the monitoring plan as outlined in the registered PDD ^{/01/} and monitoring methodology ^{/08/} .
	Monitoring equipment	The meter of 0.2s accuracy class is used. (Calibration details of meter is provided separately in this section, under the heading "Calibration of meters")
	Is the accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	The accuracy of the monitoring equipment used to measure the values is 0.2s, which is as per registered PDD ^{/01/} .
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	Yes. Accuracy is valid for the entire measuring range.
	Calibration frequency /interval:	The meters are calibrated every 3 years which is done in line with the procedures mentioned in the PD.
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring	Yes, the calibration interval of meter is 3 years, which is in line with the monitoring plan ^{/01/} .

	plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed, or as per the manufacturer's specifications?	
	Is the calibration of measuring equipment carried out by an accredited person or institution?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is(are) calibration(s) valid for the whole reporting period?	Yes, the calibration is valid for the entire monitoring period.
	How were the values in the monitoring report verified?	The values in the monitoring report were verified from the monthly JMR and share certificates ^{/09/} . Value monitored: 176,250.06 MWh
	If applicable, has the reported data been cross-checked with other available data?	The monthly reported values of $EG_{pj,y}$ were further cross-checked with share certificates issued by state utility ^{/09/} .
	Does the data management ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data ensure correct transfer of data and reporting of emission reduction management. QA/QC processes are in place.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by	NA

	Appendix 1 to the CDM Project Standard?	
Findings	An issue in CL#03 was raised for data mismatch with sources and was successfully resolved on correction by PP.	
Conclusion	The parameter has been monitored appropriately in accordance with the registered monitoring plan^{/01/} and applied methodology^{/08/}. The monitored data were recorded consistently as per the approved frequency in the monitoring plan. Since 100% of data has been monitored and verified, the verification team can ascertain that the values used for the calculation of emission reduction are free from material errors.	

2. Net Electricity export recorded at 0.2 accuracy class GETCO meter at the EIL, EG_{GETCO, Export}.

Means of verification	Criteria/Requirements	Assessment/Observation
	Measuring/Reading/Recording frequency	The parameter for Export is continuously measured by the electricity meters installed at EIL Bhogat Substation.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The reporting frequency is monthly basis and is in line with the monitoring plan as outlined in the registered PDD ^{/01/} and monitoring methodology ^{/08/} .
	Monitoring equipment	Energy meters of accuracy class 0.2s are used, (Calibration details of meter is provided separately in this section, under the heading "Calibration of meters")
	Is the accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with	The accuracy of the monitoring equipment used to measure the input values used to calculate EG _{GETCO, export} is 0.2s, which is as per the registered PDD ^{/01/} which is

	local/national standards, or as per the manufacturer's specification?	as per the norm defined in the PPA.
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	Yes. The accuracy of monitoring equipment is valid for the entire range.
	Calibration frequency /interval:	The meter calibration frequency is once every 3 years which is in line with PPA and registered monitoring plan ^{01/} .
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed, or as per the manufacturer's specifications?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is the calibration of measuring equipment carried out by an accredited person or institution?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is(are) calibration(s) valid for the whole reporting period?	Yes, the calibration is valid for the entire monitoring period.
	How were the values in the monitoring report verified?	NA
	If applicable, has the reported data been cross-checked with other available data?	NA
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data ensure correct transfer of data and reporting of emission reductions management. QA/QC processes are in place.

	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	NA
Findings	No findings were raised related to this parameter.	
Conclusion	The parameter has been monitored appropriately in accordance with the registered monitoring plan^{/01/} and applied methodology^{/08/}. The monitored data was recorded consistently as per the approved frequency in monitoring plan. Since 100% data has been monitored and verified, the verification team can ascertain that the values used for calculation of emission reduction are free from material errors.	

3. Net Electricity import recorded at 0.2 accuracy class GETCO meter at the EIL Bhogat station, EG_{GETCO, Import}:

Means of verification	Criteria/Requirements	Assessment/Observation
	Measuring/Reading/Recording frequency	The parameter for import is continuously measured by the electricity meters installed at EIL Bhogat Substation.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The reporting frequency is monthly basis and is in line with the monitoring plan as outlined in the registered PDD ^{/01/} and monitoring methodology ^{/08/} .
	Monitoring equipment	Energy meters of accuracy class 0.2s are used, (Calibration details of meter is provided separately in this section, under the heading "Calibration of meters")
	Is the accuracy of the monitoring equipment as stated in the monitoring	The accuracy of the monitoring equipment used to measure the

	plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	input values used to calculate $EG_{GETCO, Import}$ is 0.2s, which is as per the registered PDD ^{01/} which is as per the norm defined in the PPA.
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	Yes. The accuracy of monitoring equipment is valid for the entire range.
	Calibration frequency /interval:	The meter calibration frequency is once every 3 years which is in line with PPA and registered monitoring plan ^{01/} .
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed, or as per the manufacturer's specifications?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is the calibration of measuring equipment carried out by an accredited person or institution?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is(are) calibration(s) valid for the whole reporting period?	Yes, the calibration is valid for entire monitoring period.
	How were the values in the monitoring report verified?	NA
	If applicable, has the reported data been cross-checked with other available data?	NA

	Does the data management ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data ensure correct transfer of data and reporting of emission reductions management. QA/QC processes are in place.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	NA
Findings	No findings were raised related to this parameter.	
Conclusion	The parameter has been monitored appropriately in accordance with the registered monitoring plan^{/01/} and applied methodology^{/08/}. The monitored data were recorded consistently as per the approved frequency in the monitoring plan. Since 100% of data has been monitored and verified, the verification team can ascertain that the values used for the calculation of emission reduction are free from material errors.	

4. Electricity import recorded at the 0.2 accuracy class cluster meter at the Vacuum Circuit Breaker (VCB) yard, EG_{cluster meter export}

Electricity imported recorded at the 0.2 accuracy class cluster meter at the Vacuum Circuit Breaker (VCB) yard, EG_{cluster meter, Import}

Means of verification	Criteria/Requirements	Assessment/Observation
	Measuring/Reading/Recording frequency	The parameter is continually measured by EIL Bhogat substation and is provided to GEDA on monthly basis.
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. The reporting frequency is monthly basis and is in line with the monitoring plan as outlined in the registered PDD ^{/01/} and monitoring methodology ^{/08/} .

	Monitoring equipment	Energy meters of accuracy class 0.2s are used.
	Is the accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	The accuracy of the monitoring equipment used to measure the input values used to calculate $EG_{\text{cluster meter, export}}$ and $EG_{\text{cluster meter, import}}$ is 0.2s, which is as per the registered PDD ^{/01/} which is as per the norm defined in the PPA.
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	Yes. The accuracy of monitoring equipment is valid for the entire range.
	Calibration frequency /interval:	The meter calibration frequency is once every 3 years which is in line with PPA and registered monitoring plan ^{/01/} .
	Is the calibration interval in line with the monitoring plan and/or methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed, or as per the manufacturer's specifications?	Yes the calibration interval is in line with the monitoring plan. The checking and testing procedure are in preview of state utility. The PP has no involvement in it.
	Is the calibration of measuring equipment carried out by an accredited person or institution?	The meter testing and checking procedures are solely within the purview of GETCO (Gujarat Energy Transmission Corporation Limited).
	Is(are) calibration(s) valid for the whole reporting period?	Yes, the calibration is valid for the entire monitoring period.
	How were the values in the monitoring report verified?	NA

	If applicable, has the reported data been cross-checked with other available data?	NA
	Does the data management ensure the correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the data ensure correct transfer of data and reporting of emission reduction management. QA/QC processes are in place.
	In case project participants have temporarily not monitored the parameter, has either i) a deviation been approved by the CDM EB or ii) has the parameter been estimated as stipulated by Appendix 1 to the CDM Project Standard?	NA
Findings	No findings were raised related to this parameter.	
Conclusion	The parameter has been monitored appropriately in accordance with the registered monitoring plan^{/01/} and applied methodology^{/08/}. The monitored data were recorded consistently as per the approved frequency in the monitoring plan. Since 100% of data has been monitored and verified, the verification team can ascertain that the values used for the calculation of emission reduction are free from material errors.	

Parameters fixed ex-ante:

EF_{OM,y}: The parameter is described as the operating margin CO₂ emission factor of the Indian Grid and is measured in unit tCO_{2e}/MWh. The value provided in MR i.e., 1.0049 has been verified from registered PDD^{/01/} and CO₂ Baseline Database for Indian Power Sector by Central Electricity Authority^{/16/}.

EF_{grid, BM,y}: The parameter is described as the build margin CO₂ emission factor of the Indian grid which is measured in units tCO_{2e}/MWh. The value provided in MR i.e., 0.6752 has been verified from registered PDD^{/01/} and CO₂ Baseline Database for Indian Power Sector by Central Electricity Authority^{/16/}.

EF_{grid, CM,y}: The parameter is described as the combined margin CO₂ emission factor of the Indian grid which is measured in units tCO_{2e}/MWh. The value provided in MR i.e., 0.9225 has been verified from PD^{/01/}.

Calibration of meters:

The energy meters installed at the sub-station are of accuracy class 0.2s as verified through the calibration certificates^{/11/}. The installation of the meters was checked through Calibration certificates and photographs and found to be satisfactory. Details of the meters are provided below the table:

Substation / GETCO meter details:

Meter Serial No.	Make and Accuracy class	Date of the previous calibration	Date of latest calibration	Remark
GJ-2413A	Secure 0.2s	13/06/2018	13/06/2021	There is no delay in calibration
GJ-2438A	Secure 0.2s	13/06/2018	13/06/2021	There is no delay in calibration
GJ-2439A	Secure 0.2s	13/06/2018	13/06/2021	There is no delay in calibration

Cluster meter details:

Meter No.	Serial	Make and Accuracy class	Date of the previous calibration	Date of latest calibration	Remark
GJU07452		Secure 0.2	14/07/2014	14/06/2021	There is a delay in calibration. As the calibration is not in control of PP.
GJU60938		Secure 0.2	19/01/2018	14/06/2021	There is no delay.
18016929		Secure 0.2	17/06/2018	14/06/2021	There is no delay.

Given the above discussion the assessment team is able to confirm that evidence used to determine the GHG reductions and removals are sufficient and appropriate concerning quality and quantity.

GHG Calculations:

The emission reduction as per the applied methodology equals the baseline emissions (project emissions and leakage emissions for such project activities are considered zero). The formula provided for the calculation of baseline emissions is

$$ER_y = BE_y - PE_y$$

Where,

BE_y = Baseline Emissions in year y (t CO_{2e}/yr)

PE_y = Project Emissions in year y (t CO_{2e}/yr)

For calculation of baseline emissions:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where,

BE_y = Baseline Emissions in year y (t CO_{2e}/yr)

$EG_{PJ,y}$ = Quantity of net electricity supplied to the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = CO₂ emission factor of the grid in year y (t CO₂/yr)

Since the project activity is the installation of a new grid-connected renewable power plant the $EG_{PJ,y}$ is calculated as :

$$EG_{PJ,y} = EG_{facility,y}$$

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

$$EG_{PJ,y} = 176,250.06 \text{ MWh}$$

$$EF_{grid,CM,y} = 0.9225 \text{ tCO}_2/\text{MWh}$$

$$BE_y = 176,250.06 * 0.9225 = 162,589 \text{ tCO}_2e$$

$$BE_y = 162,589 \text{ tCO}_2e$$

The emission reduction as

$$ER_y = BE_y - PE_y$$

$$PE_y = 0$$

Thus, $ER_y = BE_y$

$$ER_y = 162,589 \text{ tCO}_2e$$

The verification team confirms that appropriate methods and formulae for calculating baseline emissions have been followed. The assumptions, emission factors, and default values that were applied in the calculations are justified. The actual emission reduction achieved during the current monitoring period is 26.90%% lower than the estimated amount of emission reduction as determined in the registered PDD^{01/} which is due to the low PLF achieved by the project activity during the monitoring period. The generation depends on wind availability and other climatic conditions which is acceptable as it is not in the control of the project participant.

All the data were made available and been monitored as per the required monitoring frequency. The means of verification for the values of parameters, used for baseline emissions calculations are described above.

CAR#05 was raised regarding a transparent description of calibration frequency and it was successfully resolved.

4.6 Non-Permanence Risk Analysis

Not applicable for the project activity.

5 VERIFICATION CONCLUSION

Earthood Services Private Limited (ESPL) contracted by Tata Power Renewable Energy Limited, to perform the independent verification of the emission reductions for the VCS project activity 'AES Saurashtra Windfarms' (VCS ID- 1039) in India for the monitoring period 01/09/2018 to 06/06/2021 as reported in monitoring report version 03.3 dated 08/11/2022. Tata Power Renewable Energy Limited is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emission reduction from the project activity.

It is our responsibility to express an independent verification statement on the reported GHG emission reductions from the project activity. ESPL commenced the verification based on the baseline and monitoring methodology ACM0002 Version 12.3.0, the monitoring plan contained in the registered PD^{/01/} and MR^{/02/}.

ESPL verification approach is based on the understanding of the risks associated with reporting GHG emission data and the controls in place to mitigate these. ESPL planned and performed the verification by obtaining evidence and other information and explanations that ESPL considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion, the GHG emissions reductions reported for the project activity for the period 01/09/2018 to 06/06/2021 are fairly stated in the Monitoring Report^{/02/}. The GHG emission reductions were calculated correctly based on the approved baseline and monitoring methodology ACM0002, version 12.3.0^{/08/} and VCS standard version 4.3^{/05/}.

Verification period: From 01/09/2018 to 06/06/2021

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2018	17,324	0	0	17,324
2019	64,592	0	0	64,592
2020	57,591	0	0	57,591
2021	23,082	0	0	23,082
Total	162,589	0	0	162,589

The actual emission reductions for this monitoring report are 162,589 tCO₂e whereas the estimated emission reduction for this monitoring period is 222,435 tCO₂e as per the registered PD. Thus, the

emission reduction achieved during the current monitoring period is 26.90%% lower than the estimated amount.



Approved by

Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 09/11/2022

Place: Gurgaon, Haryana

APPENDIX 1: DOCUMENT REFERENCES

S. No.	Title of documents	Version	Date
1.	Registered VCS PD https://registry.verra.org/app/projectDetail/VCS/1039	03.1	21/03/2013
	Registered CDM PDD https://registry.verra.org/app/projectDetail/VCS/1039	4.0	06/01/2012
2.	Monitoring Report	03.1	19/06/2022
	Final monitoring report	03.3	08/11/2022
3.	ER sheet	-	19/06/2022
4.	Validation Reports for the project activity Available at Verra registry for project activity page: https://registry.verra.org/app/projectDetail/VCS/1039		14/03/2011
5.	VCS standard https://verra.org/wp-content/uploads/2022/06/VCS-Standard_v4.3.pdf	4.3	-
6.	VCS program guide https://verra.org/wp-content/uploads/2022/06/VCS-Program-Guide-v4.2.pdf	4.2	-
7.	Letter of declaration from PP regarding not having created or sought any other form of environmental credits for the same period	-	-
8.	Approved Methodology ACM0002 – ‘Consolidated baseline methodology for grid-connected electricity generation from renewable sources https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQ_C0PIWPGWDN8ED5PG	12.3.0	-

S. No.	Title of documents	Version	Date
9.	Monthly JMR and share certificates	-	-
10.	Remote site audit and interviews were conducted during the project monitoring period Latest photographs of the SCADA system, WTGs, transformers, and energy meters installed at the site	-	18/07/2022
11.	Calibration certificates for the GETCO meters	-	13/06/2018
	Calibration certificates for the cluster meters		19/01/2018
			17/06/2018
12.	Excel sheet with Operations shutdown details	-	-
13.	VCS webpage for the project, VCS ID 1039; https://registry.verra.org/app/projectDetail/VCS/1039		
14.	Tool to calculate the emission factor for an electricity system	2.2.1, EB 63	
15.	Tool for the demonstrations and assessment of additionality	5.2.1, EB 39	
16.	CO2 Baseline Database for Indian Power sector https://cea.nic.in/	5	
17	CDM web page link for project activity, Project ID – 5777 https://cdm.unfccc.int/Projects/DB/DNV-CUK1328700673.83/view		

APPENDIX 2: ABBREVIATIONS

Abbreviations	Full texts
ABT	Availability Based Tariff
BEF	Baseline Emission Factor
BM	Build Margin
CAR	Corrective Action Request
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CL	Clarification Request
CMS	Central Monitoring System
CMP	Conference of Parties Serving as Meeting of Parties
CO ₂	Carbon dioxide
DISCOM	Distribution Company
EB	Executive Board
EPC	Engineering, Procurement, and Construction
ESPL	Earthood Services Private Limited
FAR	Forward Action Request
GEDA	Gujarat Energy Development Agency
GETCO	Gujarat Energy Transmission Corporation Limited
GUVNL	Gujarat Urja Vikas Nigam Limited
GHG	Green House Gas
ISO	International Standards Organization
JMR	Joint Meter Reading
kW	Kilowatt
kWh	Kilowatt-hour
MFR	Multi-Function Relay

Abbreviations	Full texts
MR	Monitoring Report
MW	Megawatt
MWh	Megawatt-hour
NEWNE	Northern Eastern Western Northern-Eastern
OandM	Operation and Maintenance
PD	Project Description
PLF	Plant Load Factor
PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance and Quality Control
SLDC	State Load Dispatch Centre
TOD	Time of the Day
UNFCCC	United Nations Framework Convention on Climate Change
VCUs	Voluntary Carbon Units
VCS	Voluntary Carbon Standard
VCSA	Voluntary Carbon Standard Association
VCS PD	VCS Project Description
WEC	Wind Energy Converter
WEG	Wind Energy Generator
WTG	Wind Turbine Generator
WWIL	Wind World (India) Ltd.

APPENDIX 3: FINDINGS OVERVIEW

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	XX	Section no.	E.2	Date : DD/MM/YYYY
Description of FAR				
Project participant response				Date : DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

Table 2. CL from this verification

CL I	01	Section no.	1.9 of MR	Date: 12/07/2022
Description of CL				
The PP is requested to submit a declaration confirming that the GHG Emissions reductions or removals generated by the project activity will not be used for compliance with an emission trading program or to meet binding limits on GHG Emissions.				
Project participant response				Date: 25/07/2022
Attached herewith is the declaration by PP confirming that the GHG Emission reductions or removals generated by the project activity will not be used for compliance with an emission trading program or to meet binding limits on GHG Emissions				
Documentation provided by project participant				
Copy of Declaration				
DOE assessment				Date: 16/08/2022
A declaration has been submitted by the PP that Emission reductions for the monitoring period, shall not be used by any other program or obligations. This is found to be satisfactory. CL #01 is closed.				

CL ID	02	Section no.	2.2 of MR	Date: 12/07/2022
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Description of CL	
1. In accordance with VCS standard version 4.3, section 3.17.4, the project proponent shall establish mechanisms for ongoing communication with local stakeholders to allow stakeholders to raise concerns about potential negative impacts during project implementation. - The PP is requested to clarify how section 2.2 of the monitoring report complies with VCS standard V4.3 requirements pertaining to the ongoing communication. 2. Details/confirmation regarding the operation of the project activity during the monitoring period i.e. breakdown/technical performance is not discussed in the MR. Please clarify	
Project participant response	Date: 25/07/2022
1. Kindly refer to section 2.2 of the revised MR for the mechanisms established by the PP for ongoing communication with local stakeholders to allow stakeholders to raise concerns about potential negative impacts during project implementation 2. Breakdown Details during the operation of the project activity are attached herewith in the excel sheets. And also, they are mentioned in the Appendix of the revised MR.	
Documentation provided by project participant	
Revised MR Breakdown Details in Excel Sheet	
DOE assessment	Date: 16/08/2022
1. The information regarding the ongoing communication with local stakeholders has been added in section 2.2 of the MR by PP. This is found to be satisfactory. 2. The information regarding the breakdown details has been provided in the MR and a detailed Excel sheet has been provided by the PP. This is found to be satisfactory.	
CL#02 is closed.	

CL ID	03	Section no.		Date : 12/07/2022
Description of CL				
1. The PP is requested to clarify the Crediting period date as it is not matching with the registered PD as well as the Verra website. 2. The PP is also requested to clarify the status of subsequent monitoring periods as information regarding previous monitoring periods is yet not updated on the Verra website. 3. Please Clarify if registration of the Project with CDM is withdrawn.				
Project participant response				Date : 25/07/2022

1. Crediting period for this project activity is mentioned as 07-June-2011 to 29-February-2012 in registered PD as well as Verra Website also and the same needs to be revised now. The actual/correct crediting period is mentioned below. Hence, the following are the respective crediting periods for this project activity VCS Crediting Period – 07-June-2011 to 06-June-2021 (Based on the Start date of project activity) CDM Crediting Period – 01-March-2012 to 28-February-2022 (Fixed) The project issued carbon credits as follows 07-June-2011 to 29-February-2012 – VCS 01-March-2012 to 31-August-2012 – CDM 01-September-2012 to 31-August-2018 – VCS (Verification is in progress and will be completed by the end of September 2022) 01-September-2018 to 06-June-2021 – VCS (with this verification, VCS crediting period will be completed) 2. Kindly refer to the above response at point 1 3. No, the project is not withdrawn from CDM. It also got the fixed crediting period of 10 years in CDM which ended on 28-February-2022	
Documentation provided by project participant	
No documentation is required	
DOE assessment	Date: 16/08/2022
The information provided by the PP is found to be satisfactory and in line with the VCS Standard V4.3, section 3.8.7 “Projects registered under other GHG programs are not eligible for VCU issuance beyond the end of the total project crediting period under those programs”. Hence, found to be satisfactory. CL#03 Closed.	

Table 3. CAR from this verification

CAR ID	04	Section no.	MR and ER sheet	Date: 12/07/2022
Description of CAR				
- The value of $EF_{grid\ BM,y}$ mentioned in the MR is incorrect. PP is requested to revise the same. - The data reported for electricity generation in ER is inconsistent with the support provided i.e., for the 0.8 MW capacity wind project for the month Jul/19, Sept/19, Dec/20, Apr/21. The PP is requested to recheck the data. - The PP is also requested to provide the invoices for all 4- capacities for the month Oct/19.				
Project participant response				Date: 25/07/2022
- The value of $EF_{grid\ BM,y}$ is 0.6752 and the same is revised in the MR - The data for the month Jul/19, Sept/19, Dec/20, and Apr/21 is re-checked and revised in the ER sheet - PP also attached the invoices for all 4- capacities for the month Oct/19.				
Documentation provided by project participant				

1. Revised ER Sheet
2. Invoices for the month of Oct/19
DOE assessment
Date: 16/08/2022
1. The value of $EF_{grid\ BM,y}$ is now revised.
2. The PP has revised the values for the above-mentioned months, which are also found to be in line with the invoices provided. The finding is closed.
CAR#04 Closed.

CL ID	05	Section no.		Date: 12/07/2022
Description of CAR				
It is observed from MR that there are calibration delays of the energy meters during the monitoring period. PP is requested to provide the clarification regarding the same as no error correction factor to the measured data is not applied to the applicable data if calibration delays are applicable.				
Project participant response				Date: 25/07/2022
<i>There is no delay in the Calibration of meters.</i> <i>Meter calibration is carried out on 13-June-2018 and the validity is for 3 years.</i> <i>The current Monitoring period is 01-Sept-2018 to 06-June-2021.</i> <i>Hence, the entire monitoring period is covered with Calibration and error factor is not applicable.</i>				
Documentation provided by project participant				
<i>Calibration Reports</i>				
DOE assessment				Date: 16/08/2022
Calibration Certificates (Dated: 13/06/2018; Valid till 13/06/2021) has been submitted by the PP for all the meters. The calibration validity was checked, and all the meters are found to be calibrated for the entire monitoring period. Calibration information is accordingly updated in the revised MR and therefore, the finding is closed.				
CL#05 is closed.				

CL ID	06	Section no.	ER SHEET	Date: 12/07/2022
Description of CL				
In the ER sheet, For the Month of Jun/21, the generation data for a Capacity of 21.600 MW is divided by 32 and for the rest of the 3, the generation data is divided by 31. The PP is requested to provide clarification regarding the same.				
Project participant response				Date: 25/07/2022
It was a typo error and it is divided by 30 in all 4-generation data.				
Documentation provided by project participant				

DOE assessment	Date: 16/08/2022
The values in the ER sheet have been revised by the PP and they are found to be in line with the support provided, the finding is closed. CL#06 is Closed.	

Table 4. FAR from this verification

FAR ID	XX	Section No.		Date: DD/MM/YYYY
Description of FAR				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

APPENDIX 4: COMPETENCY STATEMENTS

Name	Harsh Raval		
Education	Bachelor of Engineering in Chemical Engineering Masters of Science in Environmental and Energy Engineering		
Experience	15 Years		
Field	Climate change, Environment, and waste management		
Approved Roles			
Team Leader	Yes		
Validator	Yes		
Verifier	Yes		
Methodology Expert	Yes (AMS-I.D, ACM0002)		
Local expert	Yes (INDIA)		
Financial Expert	No		
Technical Reviewer	Yes		
TA Expert	Yes		
Reviewed by	Deepika Mahala (Quality Manager)	Date	08/12/2021
Approved by	Ashok Gautam (Technical Manager)	Date	08/12/2021

Name	Muskan Chawla
Education	M.Sc. Environmental Science
Experience	NA
Field	Environmental Science
Approved Roles	
Team Leader	No
Validator	No
Verifier	No
Methodology Expert	No
Local expert	No
Financial Expert	No
Technical Reviewer	No
TA Expert	No
Trainee	Yes

Reviewed by	Deepika Mahala	Date	09/03/2022
Approved by	Ashok Gautam	Date	09/03/2022

Competence Statement			
Name	Vikas Vilasrao Bankar		
Education	Electrical Engineering Certified Energy Auditor from Bureau of Energy Efficiency (BEE), India. Certified Measurement and Verification Professional (CMPVP) from Efficiency Valuation Organization (EVO).		
Experience	16+ years		
Field	Environmental Engineering		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	NO		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	Yes (TA 1.1, 1.2, 3.1)		
Reviewed by	Shifali Guleria (Quality Manager)	Date	16/06/2022
Approved by	Deepika Mahala (Technical Manager)	Date	17/06/2022